Lab Notebook: Contact Angle Experiment

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1 Objective

The aim of this experiment is to study surface wettability by measuring the contact angle of liquid drops on various surfaces using ImageJ software.

2 Key Concepts

- Surface tension, Surface energy
- Wetting and Wettability
- Hydrophilic and Hydrophobic surfaces
- Superhydrophobic surfaces
- Surface roughness and structured/patterned surfaces
- Young's equation and its variants
- Wenzel equation and Cassie-Baxter equation

1. Surface Tension

- Surface tension is the energy required to increase the surface area of a liquid or the force per unit length acting at the surface of a liquid. It arises due to the cohesive forces between liquid molecules.
- Units: N/m (Newton per meter) or J/m^2 (Joule per square meter).
- Example: Water has a high surface tension ($\sim$72 mN/m at room temperature), which allows insects to walk on its surface.

2. Surface Energy

- Surface energy is the excess energy at the surface of a material compared to its bulk. It is a measure of the work required to create a new surface.
- Units: J/m^2.
- Solids with high surface energy tend to attract liquids (hydrophilic), while low surface energy solids repel liquids (hydrophobic).

3. Wetting and Wettability

- Wetting refers to the ability of a liquid to maintain contact with a solid surface, resulting from intermolecular interactions.
- Wettability is quantified by the **contact angle** (θ) formed between the liquid and the solid surface:
  - $\theta < 90°$: Good wetting (hydrophilic).
  - $\theta > 90°$: Poor wetting (hydrophobic).
- Wettability depends on surface energy, surface roughness, and chemical composition.

4. Hydrophilic and Hydrophobic Surfaces

- **Hydrophilic surfaces**: Have high surface energy and attract water, leading to low contact angles ($\theta < 90°$).
- **Hydrophobic surfaces**: Have low surface energy and repel water, leading to high contact angles ($\theta > 90°$).
- Applications: Hydrophilic surfaces are used in antifogging coatings, while hydrophobic surfaces are used in water-repellent coatings.

5. Superhydrophobic Surfaces

- Superhydrophobic surfaces exhibit extremely high water repellency, with contact angles greater than 150° and very low contact angle hysteresis.
- These surfaces often mimic natural structures like lotus leaves, which have micro- and nanoscale roughness.
- Applications: Self-cleaning surfaces, anti-icing coatings, and drag reduction.

6. Surface Roughness

- Surface roughness refers to the irregularities or deviations in a surface from its ideal form. It significantly affects wettability and adhesion.
- Roughness can enhance hydrophobicity or hydrophilicity depending on the surface chemistry and structure.

7. Structured/Patterned Surfaces

- These are surfaces engineered with specific micro- or nanoscale patterns to control wettability, adhesion, or other properties.
- Examples: Micro-pillars, grooves, or hierarchical structures.
- Applications: Superhydrophobic surfaces, lab-on-a-chip devices, and anti-biofouling coatings.

8. Young's Equation

- Young's equation describes the balance of forces at the three-phase contact line (solid, liquid, gas):

$$\gamma_{SV} = \gamma_{SL} + \gamma_{LV} \cos\theta$$

  where:

  - γ_{SV}: Solid-vapor surface tension.
  - γ_{SL}: Solid-liquid surface tension.
  - γ_{LV}: Liquid-vapor surface tension (surface tension of the liquid).
  - θ: Contact angle.

- It assumes an ideal, smooth, and chemically homogeneous surface.

9. Wenzel Equation

- The Wenzel equation modifies Young's equation for rough surfaces:

$$\cos\theta_W = r \cos\theta$$

  where:

  - θ_W: Apparent contact angle on a rough surface.
  - r: Roughness ratio (actual surface area divided by projected area).
  - θ: Contact angle on a smooth surface.

- Wenzel's model assumes complete wetting of the rough surface (homogeneous wetting regime).

10. Cassie-Baxter Equation

- The Cassie-Baxter equation describes wetting on a rough surface where air pockets are trapped beneath the liquid:

$$\cos\theta_{CB} = f_r \cos\theta + f_a \cos\theta_a$$

  where:

  - θ_{CB}: Apparent contact angle.
  - f_r: Fraction of the surface in contact with the liquid.
  - f_a: Fraction of the surface in contact with air ($f_a = 1 - f_r$).
  - θ_a: Contact angle of air (typically 180°).

- This model applies to heterogeneous wetting regimes, such as superhydrophobic surfaces.

Summary of Key Equations

- **Young's Equation**: Ideal smooth surfaces.
- **Wenzel Equation**: Rough surfaces with complete wetting.
- **Cassie-Baxter Equation**: Rough surfaces with air trapping.

3 Comparison of Contact Angle Measurement Plug-ins

These concepts are fundamental in designing surfaces for applications like self-cleaning coatings, microfluidics, biomedical devices, and energy-efficient materials. sectionIntroduction Contact angle measurement is crucial in surface science to evaluate the wettability of materials. ImageJ provides various plug-ins for this task, each employing different computational methods. This document compares two commonly used plug-ins: the **Contact Angle Plugin** and **DropSnake Plugin**, with particular emphasis on their performance in measuring nearly symmetric droplets.

3.1 Contact Angle Plugin (Circular Fit & elliptical fit)

Methodology:

- Uses a **circular and elliptical approximation simultaneously** to estimate the contact angle.
- The user manually selects the drop baseline and contour.
- A circle is fitted to the drop profile to determine the contact angle.

Technical Analysis:

- Assumes that the drop profile follows a **perfect circular arc followed by elliptical arc**, which simplifies calculations.
- Less sensitive to **manual errors** compared to DropSnake.
- Provides **consistent and repeatable results** for symmetric drops.

Advantages:

- Works well for **small, symmetric drops**.
- Less prone to **human error**.
- Suitable for **low-contrast images**.

Limitations:

- Assumes an **ideal circular & elliptical profile**, making it unsuitable for asymmetric drops.
- Less adaptable to rough surfaces.

3.2 DropSnake Plugin

Methodology:

- Uses an **active contour (snake) algorithm** to detect the drop boundary dynamically.
- Requires the user to manually select snake points along the droplet profile.

Technical Analysis:

- Provides more flexibility by adapting to **irregular drop shapes**.
- Less effective for **symmetric drops** due to high manual sensitivity.
- **High standard deviation** observed in repeated measurements, particularly for glycerin on rough Teflon (9 μL drop).
- **Human selection of snake points** significantly impacts accuracy, leading to **up to 4-degree variations** between left and right contact angles.

Advantages:

- Highly automated, reducing user bias in some cases.
- Suitable for **irregular, asymmetric drops**.
- Can process multiple images in batch mode.

Limitations:

- **Highly user-dependent**, as manual selection of contour points significantly influences results.
- **Inconsistent results for symmetric drops**, as seen in the glycerin-Teflon experiment.
- Requires careful tuning of parameters, making it **less straightforward** than the Contact Angle Plugin.

3.3 Experimental Observations: Symmetric Drop Analysis

For a nearly symmetric glycerin drop (9 μL) on rough Teflon:

- **DropSnake produced high standard deviations**, and the left/right angles varied by **4 degrees**, indicating measurement inconsistency.
- The **Contact Angle Plugin provided more stable and repeatable results**, as its circular fit assumption aligned better with the drop's shape.
- **DropSnake was highly sensitive to small fluctuations in snake point selection**, making it unreliable for precise symmetry-based measurements.

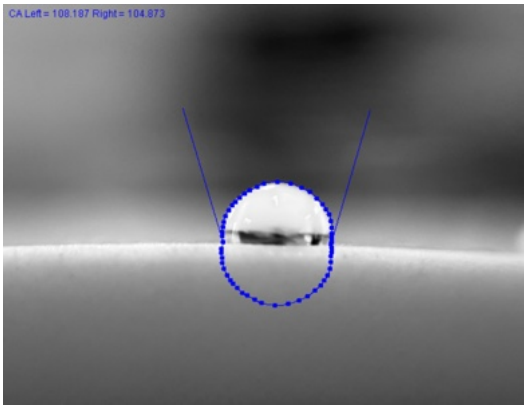


Figure 1: Teflon and glycerin contact angle measured by drop snake

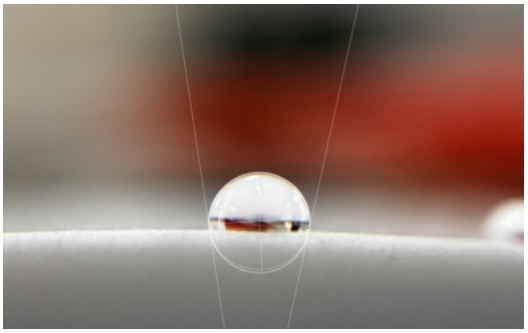


Figure 2: Teflon and glycerin contact angle measured by ContactAngle

method	Right Contact angle	Left Contact angle
DropSnake	104.9°	108.2°
ContactAngle	100.8°	100°

3.4 Conclusion

Based on the experimental analysis:

- The **Contact Angle Plugin** is **better for symmetric drops**, as it provides more stable and repeatable results.

Thus, to avoid human error and to get a consistent & error free output, I choose ContactAngle plugin for further analysis.

4 Experimental Setup

4.1 Materials & Instruments

- Micropipettes (0.5-10 μL range)
- Engineered surfaces (polished steel, PTFE, nanotextured silicon)
- Test liquids (deionized H_2O, ethylene glycol, hexadecane)
- Optical imaging system with smartphone adapter
- ImageJ v1.53 software with Contact Angle plugin

5 Experimental Procedure

5.1 Drop Generation

1. I generated drops of different volumes using micropipettes on a variety of surfaces.
2. I captured side-view images of each drop using a lens attached to my mobile phone.
3. I ensured proper contrast in the captured images.

5.2 Contact Angle Measurement

1. I uploaded the images to ImageJ software.
2. I used at least two different methods in ImageJ to estimate the contact angle.
3. I compared and discussed the advantages and limitations of each method.

5.3 Suitability of Drop Volumes

1. I identified and documented drop volumes that were unsuitable for accurate contact angle estimation.

5.4 Surface Preparation and Handling

- I ensured that the surfaces remained dry and clean.
- I only wetted the designated areas.
- I avoided touching the surfaces directly and used gloves when necessary.

5.5 Optimizing Image Quality

- I adjusted the background and lighting to achieve optimal contrast.
- I conducted trial runs with different drops to determine the best image quality.

6 Observations

6.1 Various Contact Angle for water

water	Volume(µL)	Theta C	Theta Left	Theta Right	Theta E	e
Rough Teflon						
	12	99.5	100.7	98	99.4	0.97
	4	86.8	83.5	80.3	81.9	0.91
	30	101.8	88.3	91.5	89.9	0.78
acrylic						
	4	120.4	120.3	121.1	120.7	0.98
	10	117.3	114.6	114	114.3	0.89
	0.5	118.3	116.1	116.5	116.3	0.92
copper						
	0.5	102.8	101.8	104.3	103	0.98
	1	120.5	124.7	121.1	122.9	0.87
	3	128.8	121.5	120.5	121	0.7
aluminum						
	0.6	108.6	97.7	102.9	100.3	0.81
	6	110.3	109.4	112.6	111	0.96
	3	99.3	95.2	94.9	95.1	0.91
steel						
	0.6	99.6	106.9	101.1	104	0.87
	3	95.9	92.8	89.7	91.2	0.9
	5	116.2	127.2	120.2	123.7	0.58
lotus leaf						
	1	38	35.2	35.3	35.2	0.95
	5	48	43.3	43.7	43.5	0.92
	10	47.7	40.9	41.1	41	0.89

Table 1: Table for Water and various material's Contact angle

6.1.1 Images using Water droplet

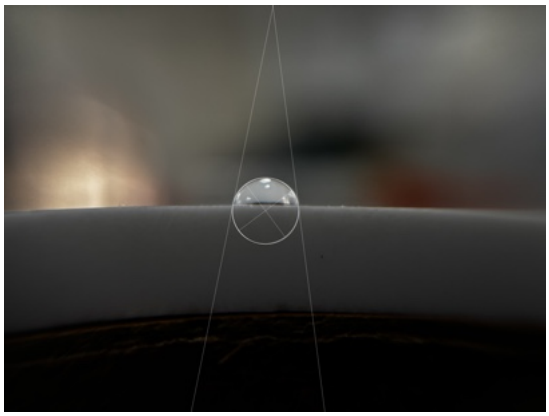


Figure 3: Teflon Rough - 12 µL

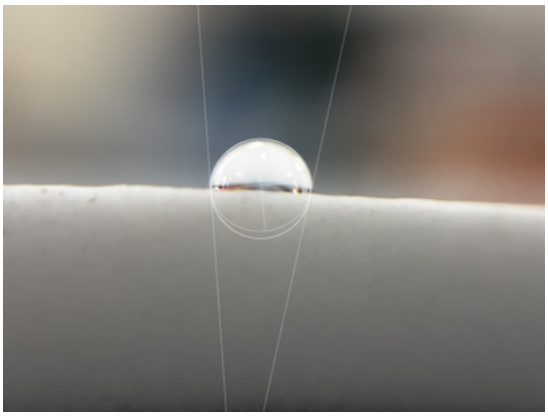


Figure 4: Teflon Rough - 4 µL

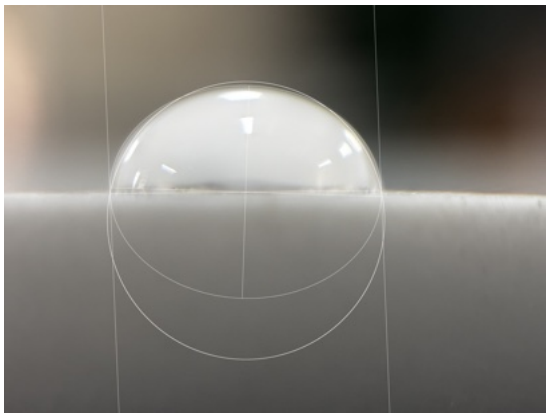


Figure 5: Teflon Rough - 30 µL

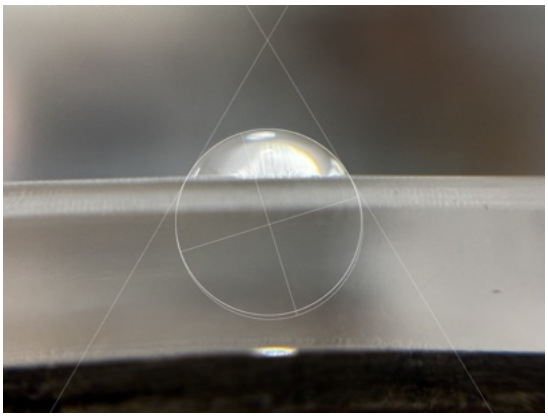


Figure 6: Acrylic - 4 µL

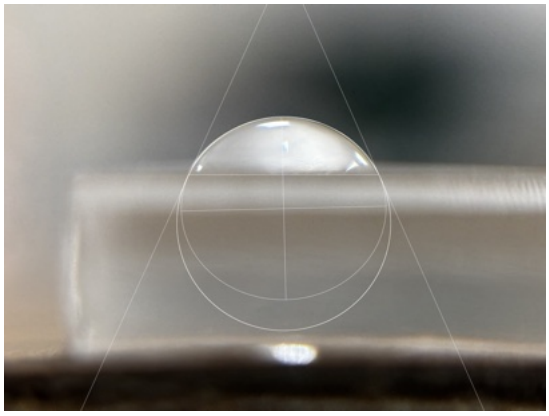


Figure 7: Acrylic - 10 µL

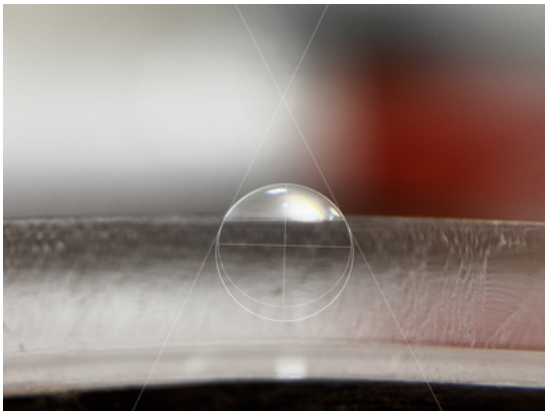


Figure 8: Acrylic - 0.5 µL



Figure 9: Copper - 0.5 µL



Figure 10: Copper - 1 µL

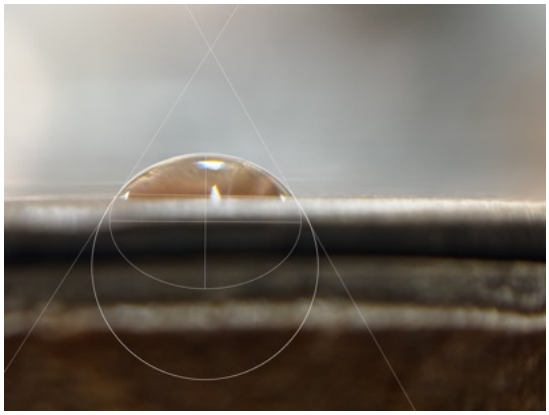


Figure 11: Copper - 3 µL

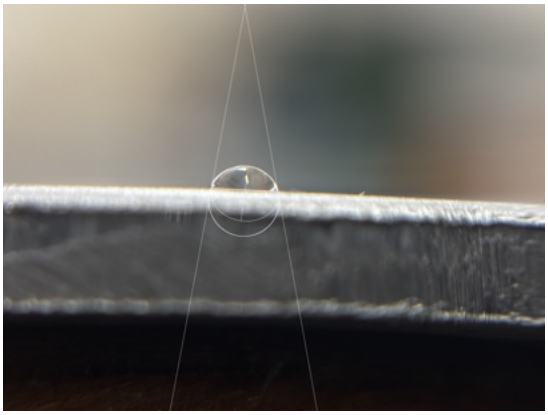


Figure 12: Aluminium - 0.6 µL

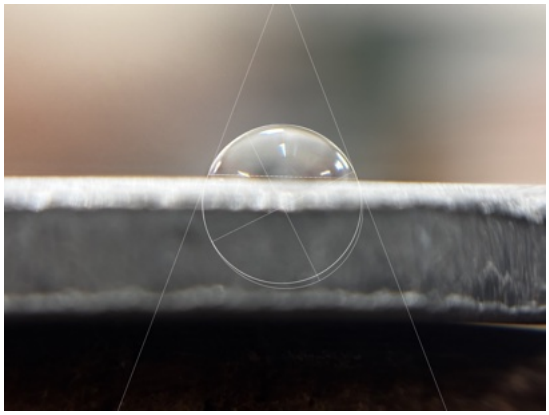


Figure 13: Aluminium - 6 µL



Figure 14: Aluminium - 3 µL

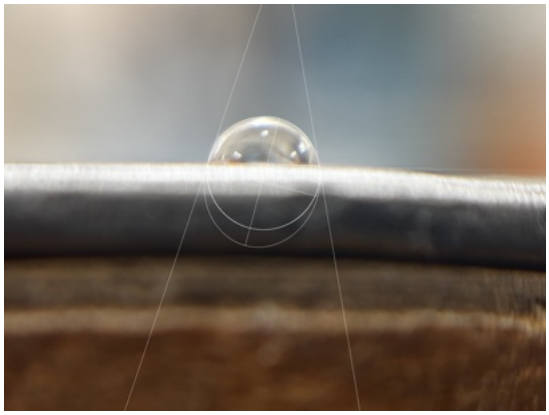


Figure 15: Steel - 0.6 µL

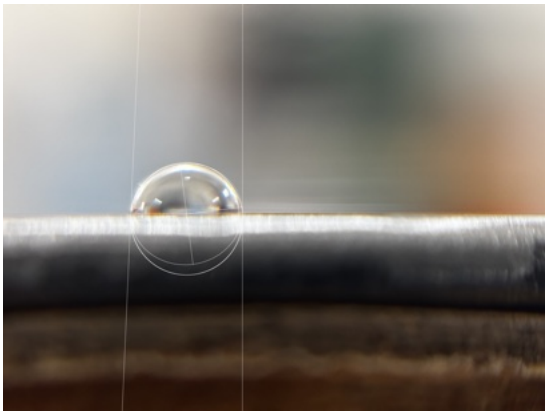


Figure 16: Steel - 3 µL

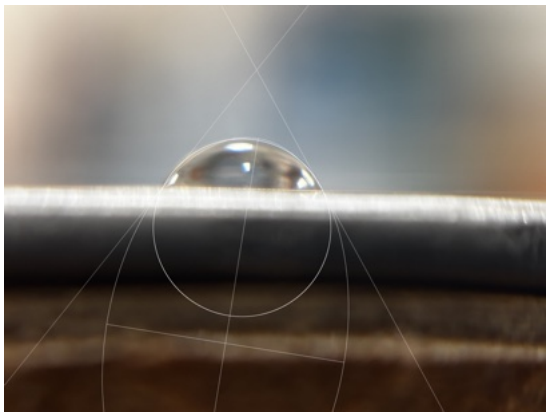


Figure 17: Steel - 5 µL

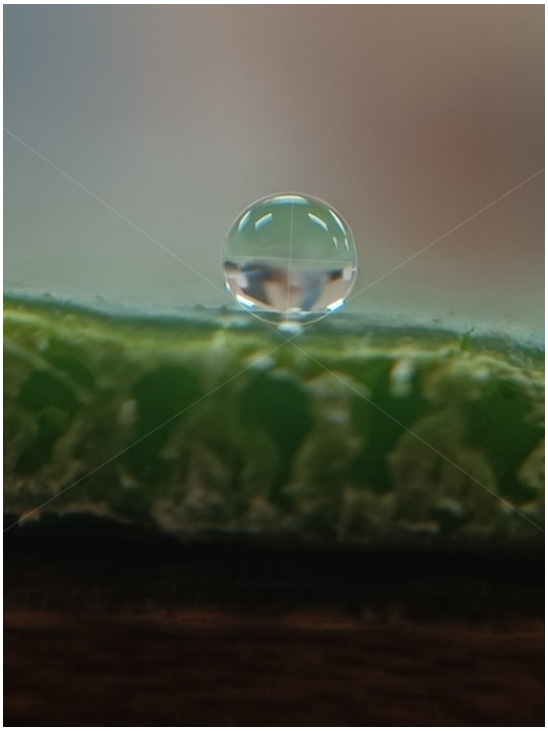


Figure 18: Lotus - 1 µL

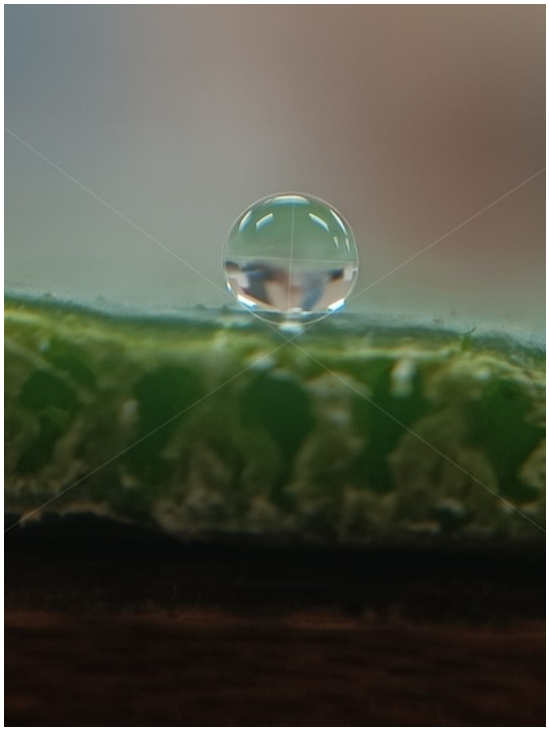


Figure 19: Lotus - 5 µL

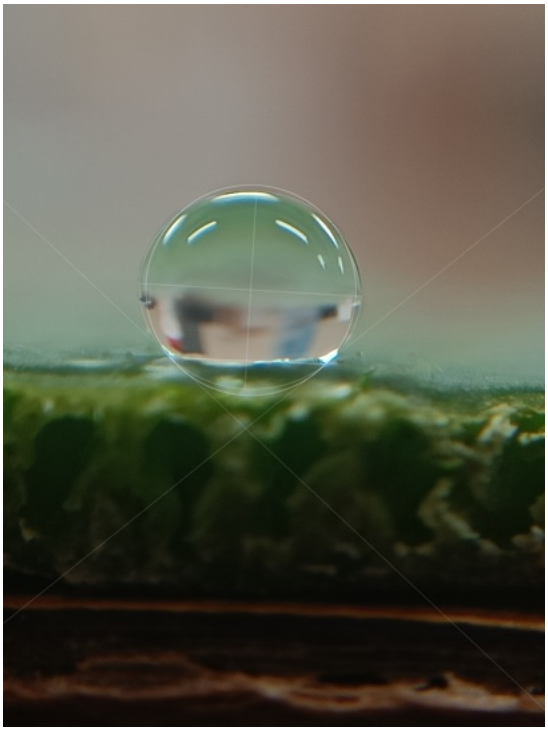


Figure 20: Lotus - 10 µL

6.2 Various Contact Angle for Glycerin

glycerin	Volume(µL)	Theta C	Theta Left	Theta Right	Theta E	e
teflon rough	5	76.4	74.4	71.6	73	0.94
	9	81.9	80	79.2	79.6	0.96
	20	101.2	86.8	88	87.4	0.77
copper	2	129.5	130.5	129.7	130.1	0.95
	5	121.6	114.7	121.8	118.2	0.85
	0.5	110.3	102.2	109.2	105.7	0.86
aluminum						
	1	107.3	107	107.2	107.1	1
	5	94.1	95.9	95	95.4	0.97
	0.5	108.4	106.3	104	105.2	0.91
smooth teflon						
	1	69	68	66.6	67.3	0.97
	4	73.7	70.8	66.7	68.8	0.92
	10	57	50.3	54.3	52.3	0.91
lotus						
	8	49.1	41.9	43.7	42.8	0.89
	5	55	49.7	51.4	50.6	0.93
	3	38.7	36.3	35.7	36	0.95
steel						
	5	89.7	80.5	80.9	80.7	0.84
	10	93.8	86.1	87.5	86.8	0.86

6.2.1 Images using glycerin droplet

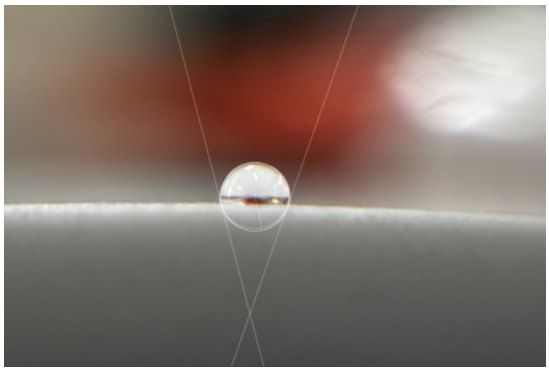


Figure 21: Teflon Rough - 5 µL Glycerin

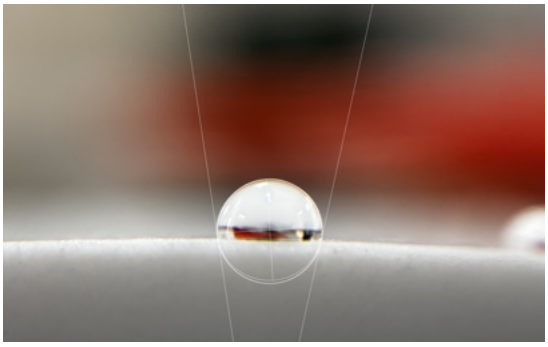


Figure 22: Teflon Rough - 9 µL Glycerin

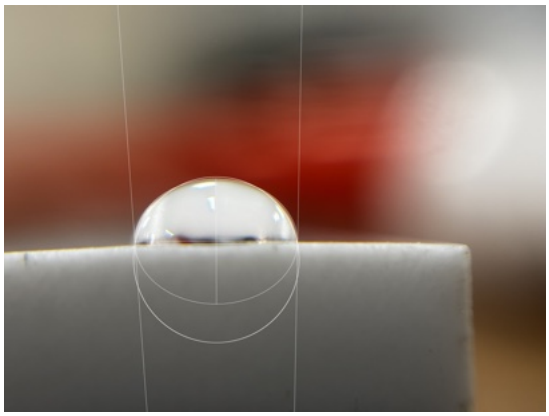


Figure 23: Teflon Rough - 20 µL Glycerin



Figure 24: Copper - 2 µL Glycerin



Figure 25: Copper - 5 µL Glycerin



Figure 26: Copper - 0.5 µL Glycerin

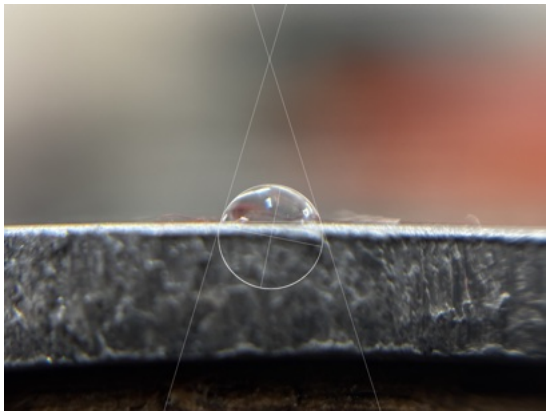


Figure 27: Aluminium - 1 µL Glycerin



Figure 28: Aluminium - 5 µL Glycerin

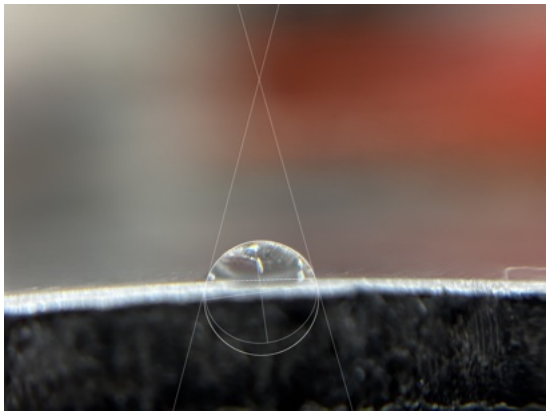


Figure 29: Aluminium - 0.5 µL Glycerin

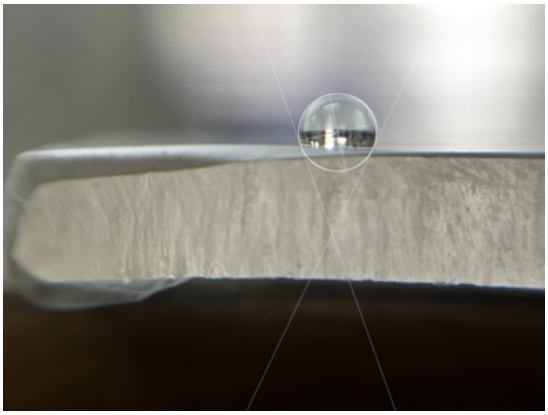


Figure 30: Smooth Teflon - 1 µL Glycerin

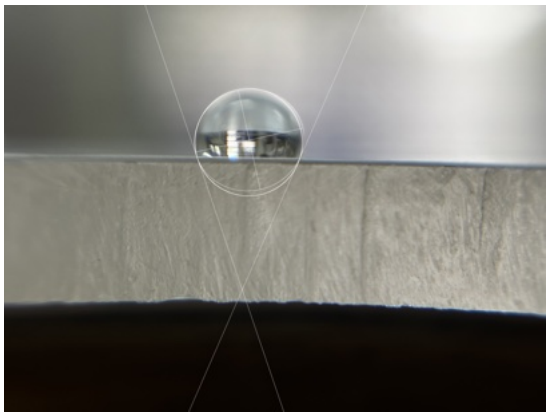


Figure 31: Smooth Teflon - 4 µL Glycerin



Figure 32: Smooth Teflon - 10 µL Glycerin



Figure 33: Lotus - 8 µL Glycerin

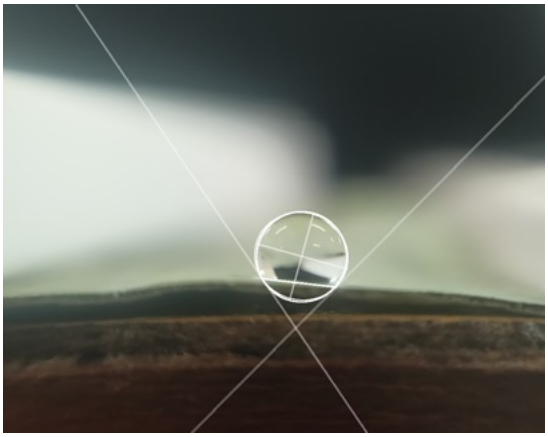


Figure 34: Lotus - 5 µL Glycerin

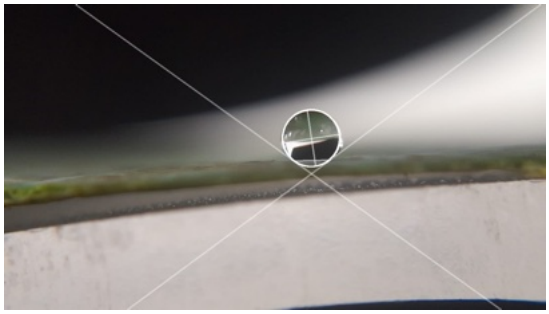


Figure 35: Lotus - 3 µL Glycerin

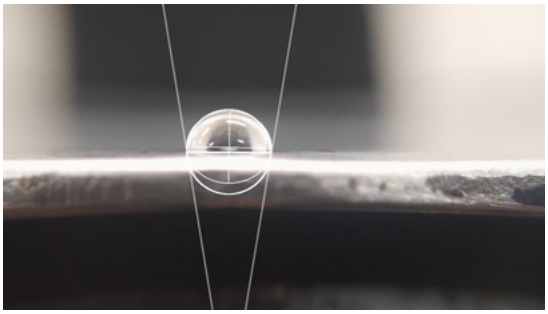


Figure 36: Steel - 5 µL Glycerin

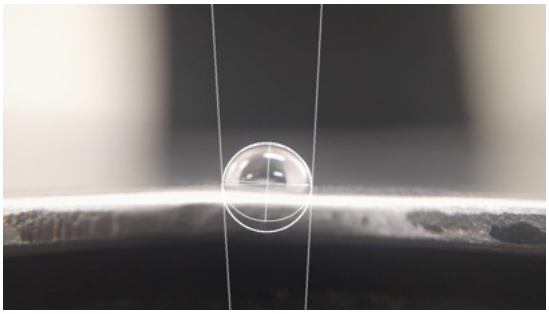


Figure 37: Steel - 10 µL Glycerin

6.3 Key terms

- **Theta C** (θ_c): The contact angle calculated using a circular approximation. This assumes that the drop's shape follows a perfect circle at the contact line, simplifying the measurement.
- **Theta E** (θ_e): The contact angle estimated using an elliptical approximation. This is useful when the drop shape deviates from a perfect circle, providing a more accurate contact angle for asymmetric or stretched drops.

- **Theta Left (θ_L) and Theta Right (θ_R)**: These are the angles formed between the baseline (solid surface) and the tangent lines at the left and right sides of the drop, respectively. These angles help determine how symmetric the drop is. If $\theta_L \neq \theta_R$, the drop may be influenced by surface heterogeneity or external forces.
- **Radius Circle**: The radius of the best-fit circular approximation used to calculate θ_c.
- **StDev (Standard Deviation)**: The statistical variation in the contact angle measurements. Higher values indicate greater uncertainty in the fitting process.
- **Ellipse StDev**: The standard deviation of the elliptical fitting process, quantifying the error in the elliptical approximation.
- **e (Eccentricity of Ellipse)**: A measure of how elongated the ellipse is. A value closer to 0 indicates a nearly circular shape, while higher values indicate a more stretched or distorted drop profile.

7 Contact Angle Determination

The ImageJ Contact Angle plugin measures the **exterior angle** (θ_{exterior}), which is the angle outside the droplet. The **interior contact angle** (θ) is calculated as:

$$\theta = 180^\circ - \theta_{\text{exterior}}$$

The elliptical approximation (θ_e) was used for asymmetric drops, and the standard deviation (σ) between the left (θ_L) and right (θ_R) angles was calculated to quantify asymmetry-induced error.

7.1 Mathematical Model for Standard Deviation

The standard deviation (σ) of the contact angle measurements is calculated to quantify the variability between the left (θ_L) and right (θ_R) angles. The formula for standard deviation is:

$$\sigma = \sqrt{\frac{(\theta_L - \bar{\theta})^2 + (\theta_R - \bar{\theta})^2}{2}}$$

where $\bar{\theta}$ is the mean of θ_L and θ_R:

$$\bar{\theta} = \frac{\theta_L + \theta_R}{2}$$

A low standard deviation ($\sigma < 2^\circ$) indicates high symmetry in the droplet shape, while a high standard deviation ($\sigma \geq 2^\circ$) suggests asymmetry due to surface heterogeneity or imaging issues.

7.2 Suitability Criteria

The suitability of a contact angle measurement is determined based on the following criteria:

- **Standard Deviation (σ)**:

  - $\sigma < 2°$: Suitable (high symmetry).
  - $\sigma \geq 2°$: Unsuitable (high asymmetry).

- **Drop Volume**:
  - Optimal range: $4\,\mu$L to $10\,\mu$L for water, $5\,\mu$L to $9\,\mu$L for glycerin.
  - Unsuitable volumes: $> 20\,\mu$L (instability) or $< 1\,\mu$L (evaporation).

- **Elliptical vs. Circular Fit**:
  - High eccentricity ($e > 0.8$): Use θ_e (elliptical fit).
  - Low eccentricity ($e \leq 0.8$): Use θ_c (circular fit).

7.3 Wettability Analysis

Wettability is determined by the contact angle (θ) of a liquid on a solid surface. The following classification is used:

- $\theta < 90°$: Hydrophilic (good wetting).
- $\theta > 90°$: Hydrophobic (poor wetting).
- $\theta > 150°$: Superhydrophobic (extremely poor wetting).

7.4 Water Droplets

The wettability of surfaces for water droplets is analyzed as follows:

- **Rough Teflon**: Contact angles ($\theta \approx 80.6°$ to $98.1°$) indicate hydrophobic behavior.
- **Acrylic**: Low contact angles ($\theta \approx 59.3°$ to $65.7°$) suggest hydrophilic behavior.
- **Copper**: Contact angles ($\theta \approx 57.1°$ to $77.0°$) indicate hydrophilic behavior, likely due to oxidation or contamination.
- **Aluminum**: Contact angles ($\theta \approx 69.0°$ to $84.9°$) suggest moderate hydrophilicity, possibly due to surface oxidation.
- **Steel**: Contact angles ($\theta \approx 56.3°$ to $88.8°$) indicate hydrophilic behavior, with some variability due to surface roughness.
- **Lotus Leaf**: High contact angles ($\theta \approx 144.8°$ to $139.0°$) confirm superhydrophobic behavior.

7.5 Glycerin Droplets

The wettability of surfaces for glycerin droplets is analyzed as follows:

- **Rough Teflon**: Contact angles ($\theta \approx 92.6°$ to $107.0°$) indicate hydrophobic behavior.
- **Copper**: Low contact angles ($\theta \approx 49.9°$ to $74.3°$) suggest hydrophilic behavior.

- **Aluminum**: Contact angles ($\theta \approx 72.9^\circ$ to 84.6°) indicate moderate hydrophilicity.
- **Smooth Teflon**: Contact angles ($\theta \approx 112.7^\circ$ to 127.7°) confirm hydrophobic behavior.
- **Steel**: Contact angles ($\theta \approx 93.2^\circ$ to 99.3°) suggest hydrophobic behavior.
- **Lotus Leaf**: High contact angles ($\theta \approx 137.2^\circ$ to 144.0°) confirm superhydrophobic behavior.

8 Results and Analysis

8.1 Water Droplets

The contact angles for water droplets on various surfaces are summarized in Table 2. Key observations include:

- **Rough Teflon**: Contact angles ($\theta \approx 80.6^\circ$ to 98.1°) indicate hydrophobic behavior.
- **Acrylic**: Low contact angles ($\theta \approx 59.3^\circ$ to 65.7°) suggest hydrophilic behavior.
- **Copper**: Contact angles ($\theta \approx 57.1^\circ$ to 77.0°) indicate hydrophilic behavior, likely due to oxidation or contamination.
- **Aluminum**: Contact angles ($\theta \approx 69.0^\circ$ to 84.9°) suggest moderate hydrophilicity, possibly due to surface oxidation.
- **Steel**: Contact angles ($\theta \approx 56.3^\circ$ to 88.8°) indicate hydrophilic behavior, with some variability due to surface roughness.
- **Lotus Leaf**: High contact angles ($\theta \approx 144.8^\circ$ to 139.0°) confirm superhydrophobic behavior.
- **Unsuitable Volumes**: Drops with volumes $> 20\,\mu$L or $< 1\,\mu$L were unstable or prone to evaporation.

Table 2: Contact Angles for Water Droplets

Surface	Volume (μL)	θ_e (Interior)	σ (θ_L, θ_R)	Suitability
Rough Teflon	12	$80.6° \pm 1.9°$	1.9°	Suitable
Rough Teflon	4	$98.1° \pm 2.3°$	2.3°	Suitable
Rough Teflon	30	$90.1° \pm 2.3°$	2.3°	Unsuitable
Acrylic	4	$59.3° \pm 0.6°$	0.6°	Suitable
Acrylic	10	$65.7° \pm 0.3°$	0.3°	Suitable
Acrylic	0.5	$63.7° \pm 0.2°$	0.2°	Suitable
Copper	0.5	$77.0° \pm 1.3°$	1.3°	Suitable
Copper	1	$57.1° \pm 1.8°$	1.8°	Suitable
Copper	3	$59.0° \pm 0.5°$	0.5°	Suitable
Aluminum	0.6	$79.7° \pm 2.6°$	2.6°	Unsuitable
Aluminum	6	$69.0° \pm 1.6°$	1.6°	Suitable
Aluminum	3	$84.9° \pm 0.2°$	0.2°	Suitable
Steel	0.6	$76.0° \pm 2.9°$	2.9°	Unsuitable
Steel	3	$88.8° \pm 1.6°$	1.6°	Suitable
Steel	5	$56.3° \pm 3.5°$	3.5°	Unsuitable
Lotus Leaf	1	$144.8° \pm 0.1°$	0.1°	Suitable
Lotus Leaf	5	$136.5° \pm 0.2°$	0.2°	Suitable
Lotus Leaf	10	$139.0° \pm 0.1°$	0.1°	Suitable

8.2 Glycerin Droplets

The contact angles for glycerin droplets are summarized in Table 3. Key observations include:

- **Rough Teflon**: Contact angles ($\theta \approx 92.6°$ to $107.0°$) indicate hydrophobic behavior.
- **Copper**: Low contact angles ($\theta \approx 49.9°$ to $74.3°$) suggest hydrophilic behavior.
- **Aluminum**: Contact angles ($\theta \approx 72.9°$ to $84.6°$) indicate moderate hydrophilicity.
- **Smooth Teflon**: Contact angles ($\theta \approx 112.7°$ to $127.7°$) confirm hydrophobic behavior.
- **Steel**: Contact angles ($\theta \approx 93.2°$ to $99.3°$) suggest hydrophobic behavior.
- **Lotus Leaf**: High contact angles ($\theta \approx 137.2°$ to $144.0°$) confirm superhydrophobic behavior.
- **Unsuitable Volumes**: Drops with volumes $> 20\,\mu$L or $< 1\,\mu$L were unstable or prone to evaporation.

Table 3: Contact Angles for Glycerin Droplets

Surface	Volume (μL)	θ_e (Interior)	σ (θ_L, θ_R)	Suitability
Rough Teflon	5	$107.0° \pm 1.4°$	1.4°	Suitable
Rough Teflon	9	$100.4° \pm 0.4°$	0.4°	Suitable
Rough Teflon	20	$92.6° \pm 0.6°$	0.6°	Suitable
Copper	2	$49.9° \pm 0.4°$	0.4°	Suitable
Copper	5	$61.8° \pm 3.6°$	3.6°	Unsuitable
Copper	0.5	$74.3° \pm 3.5°$	3.5°	Unsuitable
Aluminum	1	$72.9° \pm 0.1°$	0.1°	Suitable
Aluminum	5	$84.6° \pm 0.5°$	0.5°	Suitable
Aluminum	0.5	$74.8° \pm 1.2°$	1.2°	Suitable
Smooth Teflon	1	$112.7° \pm 0.7°$	0.7°	Suitable
Smooth Teflon	4	$111.2° \pm 2.1°$	2.1°	Unsuitable
Smooth Teflon	10	$127.7° \pm 2.0°$	2.0°	Unsuitable
Lotus Leaf	8	$137.2° \pm 0.9°$	0.9°	Suitable
Lotus Leaf	5	$132.7° \pm 1.0°$	1.0°	Suitable
Lotus Leaf	3	$144.0° \pm 0.3°$	0.3°	Suitable
Steel	5	$99.3° \pm 0.2°$	0.2°	Suitable
Steel	10	$93.2° \pm 0.7°$	0.7°	Suitable

9 Error Analysis

The primary sources of error in the experiment include:

- **Asymmetry**: Differences between θ_L and θ_R ($\sigma \geq 2°$) indicate surface heterogeneity or imaging issues.
- **Drop Volume**: Large volumes ($> 20\,\mu$L) caused spreading, while very small volumes ($< 1\,\mu$L) led to evaporation effects.
- **Elliptical vs. Circular Fit**: High eccentricity ($e > 0.8$) favored θ_e, but discrepancies between θ_c and θ_e highlighted measurement uncertainty.

10 Conclusion

The experiment successfully measured contact angles for water and glycerin droplets on various surfaces. After correcting for the exterior angle measurement, the results align with expected surface behaviors:

- Hydrophobic surfaces (e.g., Rough Teflon, Smooth Teflon) exhibited high contact angles.
- Hydrophilic surfaces (e.g., Copper, Aluminum) exhibited low contact angles.
- Superhydrophobic surfaces (e.g., Lotus Leaf) exhibited contact angles $> 150°$.